LAB GUIDE to RTL-to-GDSII

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Solution Version: DDI22.16.000-ISR6

OA Version: oa_v22.60.095

Date Revised: 18-Aug-2024

<Lab Directory Hierarchy>

- cd ~/vlsi_edu

```
.
|-- constraints
|   `-- sample.sdc
|-- lec
|   `-- sample.do
|-- lef
|   |-- gsclib045_macro.lef
|   `-- gsclib045_tech.lef
|-- lib
|   |-- fast_vdd1v0_basicCells.lib
|   |-- fast_vdd1v0_basicCells.v
|   |-- slow_vdd1v0_basicCells.lib
|   `-- slow_vdd1v0_basiccells.v
|-- pnr
|   `-- sample.view
|-- rtl
|-- simulation
`-- syn
    |-- outputs
    |-- reports
    `-- sample_synth_script.tcl
```

1. Each Verilog source you are practicing goes into "**rtl**" directory
2. Each Simulation practice are experienced in "**simulation**" directory
3. There will be so many files generating in the process of LAB, so you are carefully and wisely to managing your directory; it is advised to make each directory for you not to be confused.

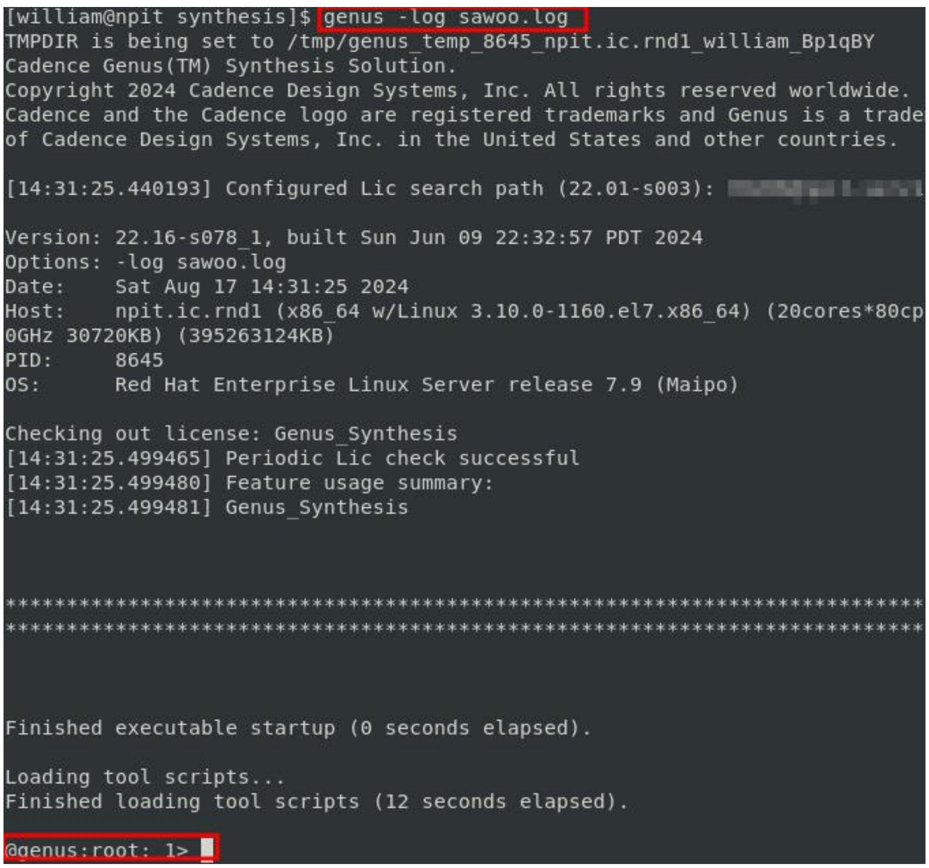
<Genus Brief Synthesizing Sequence(Without DFT)>

Environmental Setup and Invocation

- Source C-shell Environment
    - linux_shell> source ~/digital.cshrc
- Invoke Genus™ Synthesis Solution

1. Interactive Mode
    - linux_shell> genus -log {log_name}.log
2. Script Mode
    - linux_shell> genus -f {script_name}.tcl -log {log_name}.log

- After successful invocation, terminal changes as below:



```
[william@npit synthesis]$ genus -log sawoo.log
TMPDIR is being set to /tmp/genus_temp_8645_npit.ic.rnd1_william_Bp1qBY
Cadence Genus(TM) Synthesis Solution.
Copyright 2024 Cadence Design Systems, Inc. All rights reserved worldwide.
Cadence and the Cadence logo are registered trademarks and Genus is a trade
of Cadence Design Systems, Inc. in the United States and other countries.

[14:31:25.440193] Configured Lic search path (22.01-s003):

Version: 22.16-s078_1, built Sun Jun 09 22:32:57 PDT 2024
Options: -log sawoo.log
Date:    Sat Aug 17 14:31:25 2024
Host:    npit.ic.rnd1 (x86_64 w/Linux 3.10.0-1160.el7.x86_64) (20cores*80cp
0GHz 30720KB) (395263124KB)
PID:     8645
OS:      Red Hat Enterprise Linux Server release 7.9 (Maipo)

Checking out license: Genus_Synthesis
[14:31:25.499465] Periodic Lic check successful
[14:31:25.499480] Feature usage summary:
[14:31:25.499481] Genus_Synthesis



**************************************************************************
**************************************************************************



Finished executable startup (0 seconds elapsed).

Loading tool scripts...
Finished loading tool scripts (12 seconds elapsed).

@genus:root: 1>
```

Process of Brief Synthesize

1. Set the library and HDL path

- genus_shell:@> set_db init_lib_search_path {LIBRARY_PATH}

```
@genus:root: 1> set_db init_lib_search_path ../lib
  Setting attribute of root '/': 'init_lib_search_path' = ../lib
1 ../lib
```

- genus_shell:@> set_db init_hdl_search_path {HDL_PATH}

```
@genus:root: 2> set_db init_hdl_search_path ../rtl
  Setting attribute of root '/': 'init_hdl_search_path' = ../rtl
1 ../rtl
```

2. Load Library

- genus_shell:@> read_libs {LIBRARY_NAME.lib}

```
@genus:root: 4> read_libs slow_vdd1v0_basicCells.lib
```

3. Load HDL file[s]

- genus_shell:@> read_hdl {HDL_NAME.v}

```
@genus:root: 5> read_hdl counter.v
```

4. Elaboration

- genus_shell:@> elaborate

```
@genus:root: 6> elaborate
  Library has 324 usable logic and 128 usable sequential lib-cells.
Info    : Elaborating Design. [ELAB-1]
        : Elaborating top-level block 'counter' from file '../rtl/counter.v'.
Info    : Done Elaborating Design. [ELAB-3]
        : Done elaborating 'counter'.
Checking for analog nets...
Check completed for analog nets.
Checking for source RTL...
Check completed for source RTL.
Running Unified Mux Engine Tricks...
Completed Unified Mux Engine Tricks

Stage: post_elab
-------------------------------------------------------
| Trick             | Accepts | Rejects | Runtime (s) |
-------------------------------------------------------
| ume_constant_bmux |       0 |       0 |        0.00 |
| ume_merge         |       0 |       0 |        0.00 |
| ume_ssm           |       0 |       0 |        0.00 |
| ume_cse           |       0 |       0 |        0.00 |
| ume_shrink        |       0 |       0 |        0.00 |
| ume_sweep         |       0 |       0 |        0.00 |
-------------------------------------------------------
Starting optimize datapath shifters [v1.0] (stage: post_elab, startdef: counter, recur: true)
Completed optimize datapath shifters (accepts: 0, rejects: 0, runtime: 0.000s)
Starting clip mux common data inputs [v1.0] (stage: post_elab, startdef: counter, recur: true)
Completed clip mux common data inputs (accepts: 0, rejects: 0, runtime: 0.000s)
Starting clip the non-user hierarchies [v2.0] (stage: post_elab, startdef: counter, recur: true)
Completed clip the non-user hierarchies (accepts: 0, rejects: 0, runtime: 0.000s)
Starting basic netlist cleanup [v1.0] (stage: post_elab, startdef: counter, recur: true)
Completed basic netlist cleanup (accepts: 0, rejects: 0, runtime: 0.000s)

Stage: post_elab
--------------------------------------------------------------------
| Transform                    | Accepts | Rejects | Runtime (s) |
--------------------------------------------------------------------
| hlo_optimize_datapath_shifters |       0 |       0 |        0.00 |
| hlo_clip_mux_input           |       0 |       0 |        0.00 |
| hlo_clip                     |       0 |       0 |        0.00 |
| hlo_cleanup                  |       0 |       0 |        0.00 |
--------------------------------------------------------------------
UM:   flow.cputime  flow.realtime  timing.setup.tns  timing.setup.wns  snapshot
UM:*                                                                    elaborate
design:counter
```

5. Read constraints

- genus_shell:@> read_sdc {SDC_PATH/SDC_NAME.sdc}

```
@genus:root: 7> read_sdc ../constraints/constraints_top.sdc
Statistics for commands executed by read_sdc:
 "create_clock"              - successful      1 , failed      0 (runtime  0.01)
 "get_clocks"                - successful      4 , failed      0 (runtime  0.00)
 "get_ports"                 - successful      4 , failed      0 (runtime  0.01)
 "set_clock_transition"      - successful      2 , failed      0 (runtime  0.00)
 "set_clock_uncertainty"     - successful      1 , failed      0 (runtime  0.00)
 "set_input_delay"           - successful      1 , failed      0 (runtime  0.00)
 "set_output_delay"          - successful      1 , failed      0 (runtime  0.00)
read_sdc completed in 00:00:00 (hh:mm:ss)
```

6. Set effort for synthesize

- genus_shell:@> set_db syn_generic_effort medium
- genus_shell:@> set_db syn_map_effort medium
- genus_shell:@> set_db syn_opt_effort medium

```
@genus:root: 8> set_db syn_generic_effort medium
  Setting attribute of root '/': 'syn_generic_effort' = medium
1 medium
@genus:root: 9> set_db syn_map_effort medium
  Setting attribute of root '/': 'syn_map_effort' = medium
1 medium
@genus:root: 10> set_db syn_opt_effort medium
  Setting attribute of root '/': 'syn_opt_effort' = medium
1 medium
```

7. Synthesize through three steps

- genus_shell:@> syn_generic

```
@genus:root: 11> syn_generic
```

```
##>STEP                              Elapsed       WNS       TNS     Insts      Area    Memory
##>--------------------------------------------------------------------------------------------
##>G:Initial                               0         -         -        43       145       441
##>G:Setup                                 0         -         -         -         -         -
##>G:Early Clock Gating                    0         -         -         -         -         -
##>G:Launch ST                             0         -         -         -         -         -
##>G:Design Partition                      0         -         -         -         -         -
##>G:Create Partition Netlists             0         -         -         -         -         -
##>G:Init Power                            0         -         -         -         -         -
##>G:Budgeting                             0         -         -         -         -         -
##>G:Derenv-DB                             0         -         -         -         -         -
##>G:Debug Outputs                         0         -         -         -         -         -
##>G:ST loading                            0         -         -         -         -         -
##>G:Distributed                           0         -         -         -         -         -
##>G:Timer                                 0         -         -         -         -         -
##>G:Assembly                              0         -         -         -         -         -
##>G:DFT                                   0         -         -         -         -         -
##>G:Const Prop                            0         -         -        37       149       441
##>G:PostGen Opt                           0         -         -        37       149       441
##>G:Early Clock Gating Cleanup            0         -         -         -         -         -
##>G:Misc                                  2
##>--------------------------------------------------------------------------------------------
##>Total Elapsed                           2
##>============================================================================================
Info    : Done synthesizing. [SYNTH-2]
        : Done synthesizing 'counter' to generic gates.
      flow.cputime  flow.realtime  timing.setup.tns  timing.setup.wns  snapshot
UM:*                                                                   syn_gen
```

- genus_shell:@> syn_map

```
@genus:root: 13> syn_map
```

```
##>STEP                          Elapsed       WNS       TNS     Insts      Area    Memory
##>----------------------------------------------------------------------------------------
##>M:Initial                           0         -         -        37       149       830
##>M:Pre Cleanup                       0         -         -        37       149       830
##>M:Setup                             0         -         -         -         -         -
##>M:Early Clock Gating                0         -         -         -         -         -
##>M:Launch ST                         0         -         -         -         -         -
##>M:Design Partition                  0         -         -         -         -         -
##>M:Create Partition Netlists         0         -         -         -         -         -
##>M:Init Power                        0         -         -         -         -         -
##>M:Budgeting                         0         -         -         -         -         -
##>M:Derenv-DB                         0         -         -         -         -         -
##>M:Debug Outputs                     0         -         -         -         -         -
##>M:ST loading                        0         -         -         -         -         -
##>M:Distributed                       0         -         -         -         -         -
##>M:Timer                             0         -         -         -         -         -
##>M:Assembly                          0         -         -         -         -         -
##>M:DFT                               0         -         -         -         -         -
##>M:DP Operations                     2         -         -        22        72       830
##>M:Const Prop                        0      8714         0        22        72       830
##>M:Cleanup                           0      8714         0        22        72       830
##>M:MBCI                              0         -         -        22        72       830
##>M:PostGen Opt                       0         -         -         -         -         -
##>M:Early Clock Gating Cleanup        0         -         -         -         -         -
##>M:Const Gate Removal                0         -         -         -         -         -
##>M:Misc                              0
##>----------------------------------------------------------------------------------------
##>Total Elapsed                       2
##>========================================================================================
Info    : Done mapping. [SYNTH-5]
        : Done mapping 'counter'.
      flow.cputime  flow.realtime  timing.setup.tns  timing.setup.wns  snapshot
UM:*                                                                   syn_map
```

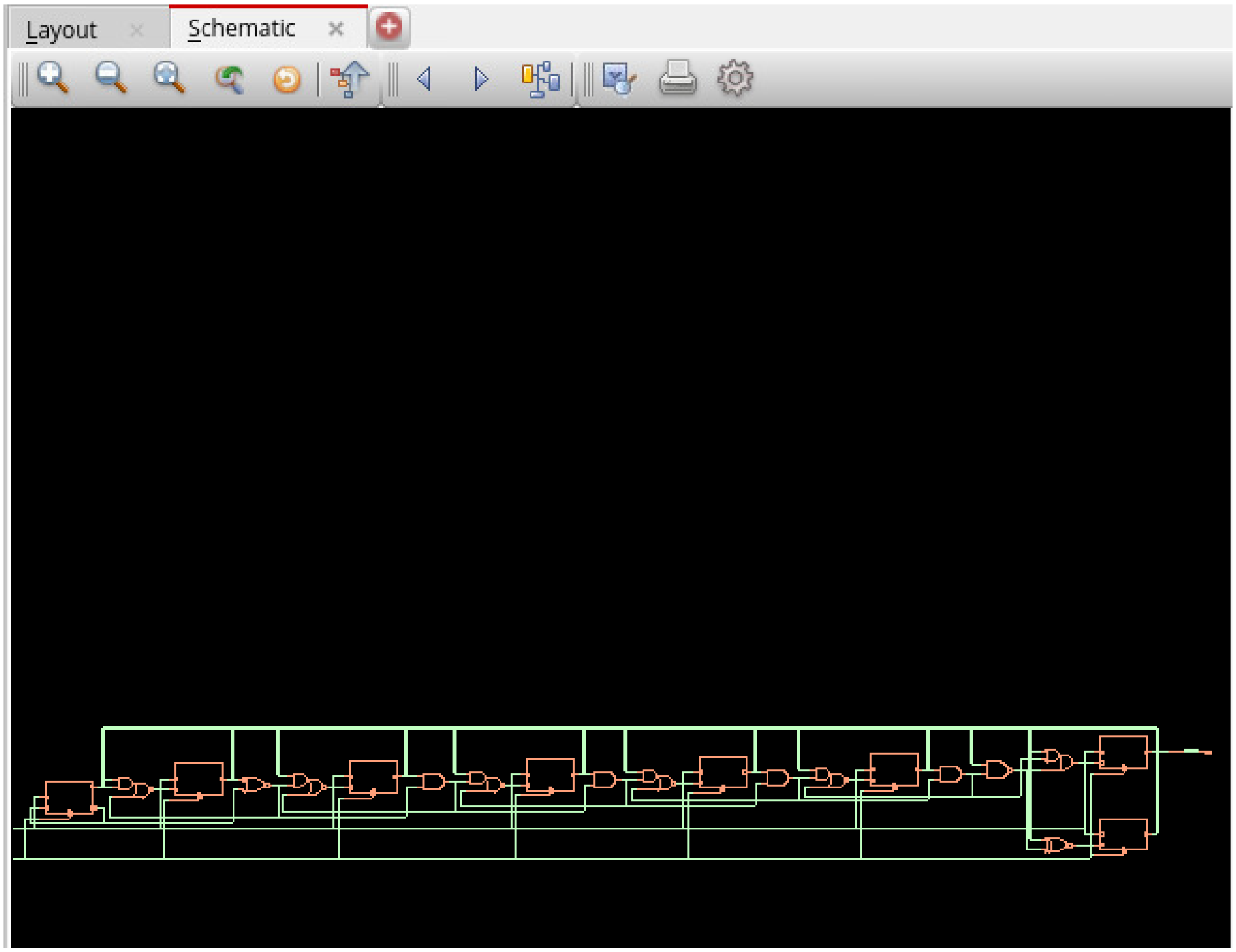
- genus_shell:@> syn_opt

```
@genus:root: 14> syn_opt
```

```
==================================
Stage : incr_opt
==================================
  =================
   Message Summary
  =================
-------------------------------------------------------------------
|  Id    |Sev |Count|               Message Text                   |
-------------------------------------------------------------------
|CFM-1   |Info|    1|Wrote dofile.                                 |
|CFM-5   |Info|    1|Wrote formal verification information.        |
|PA-7    |Info|    4|Resetting power analysis results.             |
|        |    |     |All computed switching activities are removed.|
|SYNTH-5 |Info|    1|Done mapping.                                 |
|SYNTH-7 |Info|    1|Incrementally optimizing.                     |
-------------------------------------------------------------------
Info    : Done incrementally optimizing. [SYNTH-8]
        : Done incrementally optimizing 'counter'.
      flow.cputime  flow.realtime  timing.setup.tns  timing.setup.wns  snapshot
UM:*                                                                   syn_opt
```

8. Confirm the result synthesized

```
@genus:root: 15> gui_raise
```



9. Write output files for physical implementation and verification

- genus_shell:@> write_hdl > {OUTPUT_PATH}/{DESIGN_NAME}_netlist.v

```
@genus:root: 16> write_hdl > outputs/counter_netlist.v
```

- genus_shell:@> write_sdc > {OUTPUT_PATH}/{DESIGN_NAME}_syn.sdc

```
@genus:root: 18> write_sdc > outputs/counter_syn.sdc
Finished SDC export (command execution time mm:ss (real) = 00:00).
```

- genus_shell:@> write_sdf -timescale ns -nonegchecks -recrem split -edges check_edge \
  -setuphold split > {OUTPUT_PATH}/{DESIGN_NAME}_delay.sdf

```
@genus:root: 19> write_sdf -timescale ns -nonegchecks -recrem split -edges check_edge \
==> -setuphold split > outputs/counter_delay.sdf
```

```
[william@npit outputs]$ ls -al
total 24
drwxr-xr-x. 2 william admin   79 Aug 17 15:23 .
drwxr-xr-x. 9 william admin 4096 Aug 17 14:31 ..
-rw-r--r--. 1 william admin 9218 Aug 17 15:23 counter_delay.sdf
-rw-r--r--. 1 william admin 1928 Aug 17 15:23 counter_netlist.v
-rw-r--r--. 1 william admin 1385 Aug 17 15:23 counter_syn.sdc
```

<Logical Equivalence Checking>

Environmental Setup and Invocation

- Source C-shell Environment
    - linux_shell> source ~/digital.cshrc
- Invoke Confomal™ LEC Solution

1. Interactive Mode
    - linux_shell> lec -L -nogui -color -64
2. Script Mode
    - linux_shell> lec -L -nogui -color -64 -dofile {DESIGN}.do

- After successful invocation, terminal changes as below:

```
[william@npit Equivalence_checking]$ lec -L -nogui -color -64
                          CONFORMAL (R)
                   Version 22.10-s300 (29-Aug-2022) (64 bit executable)
   Copyright (c) Cadence Design Systems, Inc., 1997-2022. All Rights Reserve


This program is proprietary and confidential information belonging to
Cadence Design Systems, Inc., and may be used and disclosed only as authoriz
in a license agreement controlling such use and disclosure.

// Warning: This version is 718 days old. You can download the latest versio
oads.cadence.com.

// Check out Conformal_Asic 22.1 license
SETUP>
```

Process of Equivalence checking

1. Set log file

- SETUP> set log file {design_lec.log} -replace

```
SETUP> set log file counter_lec.log -replace
```

2. Read library

- SETUP> read library {LIBRARY_PATH}/{LIBRARY_NAME}.lib -liberty -both

```
SETUP> read library ../lib/slow_vdd1v0_basicCells.lib -liberty -both
// Parsing file ../lib/slow_vdd1v0_basicCells.lib ...
// Warning: (RTL9.17) Cell doesn't have any output pins or all output pins are undriven (blackb
oxed) (occurrence:1)
// Note: (RTL9.26) Liberty pin with internal_node use default input_map (occurrence:16)
// Note: Read Liberty library successfully
CPU time     : 0.23    seconds
Elapse time  : 333     seconds
Memory usage : 78.70   M bytes
```

3. Read Golden Design

- SETUP> read design {RTL_DESIGN}.v -verilog -golden

```
SETUP> read design ../rtl/counter.v -verilog -golden
// Parsing file ../rtl/counter.v ...
// Golden root module is set to 'counter'
// Warning: (RTL1.5b) Potential loss of RHS msb or carry-out bit (occurrence:
// Warning: (RTL1.6) Blocking assignment is in sequential always block (occur
// Note: Read VERILOG design successfully
```

4. Read Revised Design

- SETUP> read design {GATE_LEVEL_NETLIST}.v -verilog -revised

```
SETUP> read design ../synthesis/outputs/counter_netlist.v -verilog -revised
// Parsing file ../synthesis/outputs/counter_netlist.v ...
// Revised root module is set to 'counter'
// Note: Read VERILOG design successfully
```

5. Change the mode from SETUP to LEC

- SETUP> set system mode lec

```
SETUP> set system mode lec
// Processing Golden ...
// Modeling Golden ...
// Processing Revised ...
// Modeling Revised ...
// Mapping key points ...
================================================================================
Mapped points: SYSTEM class
--------------------------------------------------------------------------------
Mapped points     PI     PO     DFF       Total
--------------------------------------------------------------------------------
Golden            2      8      8         18
--------------------------------------------------------------------------------
Revised           2      8      8         18
================================================================================
LEC>
```

6. Compare the golden with the revised netlist

- SETUP> add compare points -all
- SETUP> compare

```
LEC> add compare points -all
// 16 compared points added to compare list
LEC> compare
================================================================================
Compared points      PO     DFF       Total
--------------------------------------------------------------------------------
Equivalent           8      8         16
================================================================================
LEC>
```

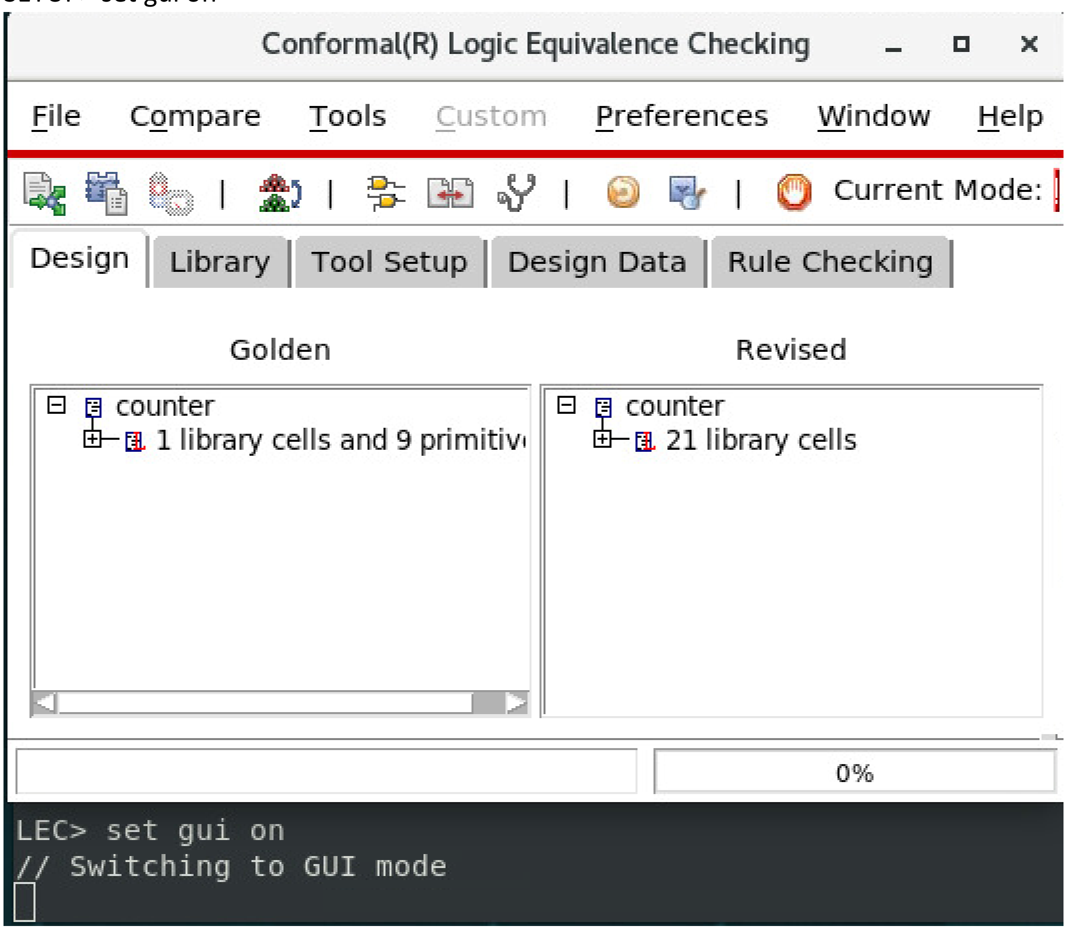
7. Generate Report

- SETUP> report verification

```
LEC> report verification
==================================================================================
                          Verification Report
----------------------------------------------------------------------------------
Category                                                                     Count
----------------------------------------------------------------------------------
1. Non-standard modeling options used:                                         0
----------------------------------------------------------------------------------
2. Incomplete verification:                                                    0
----------------------------------------------------------------------------------
3. User modification to design:                                                0
----------------------------------------------------------------------------------
4. Conformal Constraint Designer clock domain crossing checks recommended:     0
----------------------------------------------------------------------------------
5. Design ambiguity:                                                           0
----------------------------------------------------------------------------------
6. Compare Results:                                                         PASS
==================================================================================
```

8. Raise GUI

- SETUP> set gui on



<Physical Implementation>

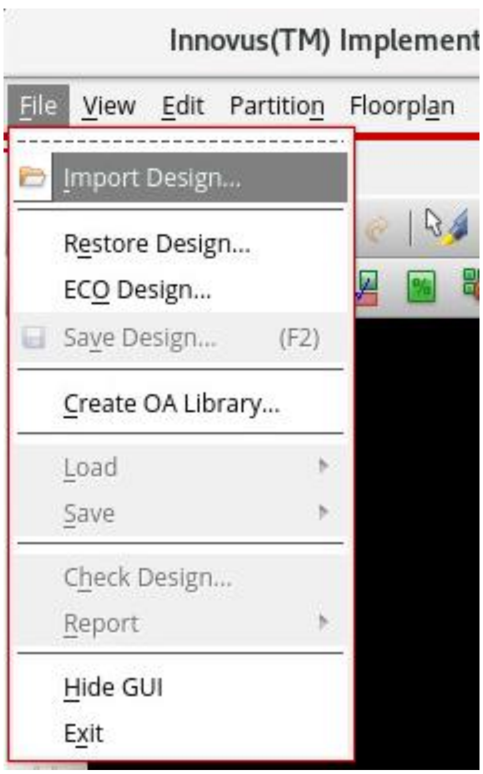
Environmental Setup and Invocation

- Source C-shell Environment
  - linux_shell> source ~/digital.cshrc
- Invoke Innovus™ Physical Implementation Solution
  1. Interactive Mode
     - linux_shell> innovus -stylus
  2. Script Mode
     - linux_shell> innovus -stylus -file {SCRIPT_FILE}.tcl

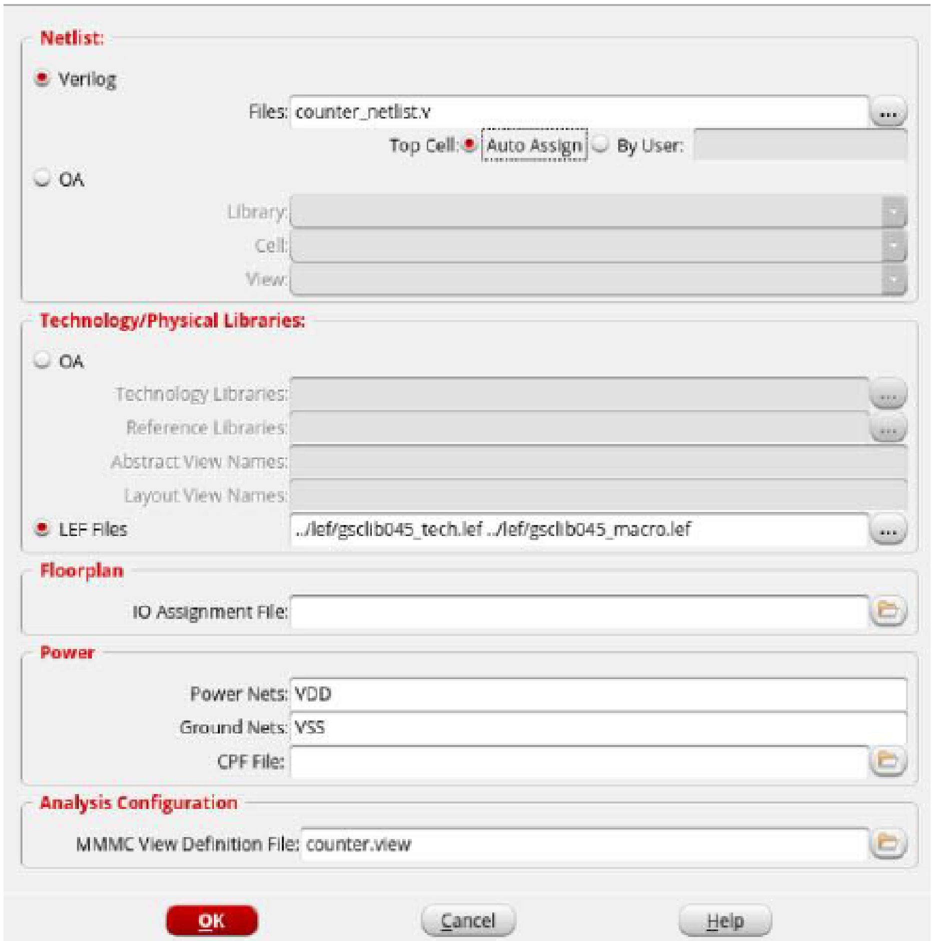
Process of Brief Physical Implementation

1. Import Design

- [File] – [Import]



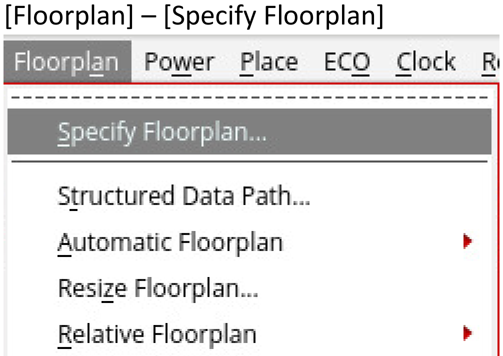
- Fill in the from as below:

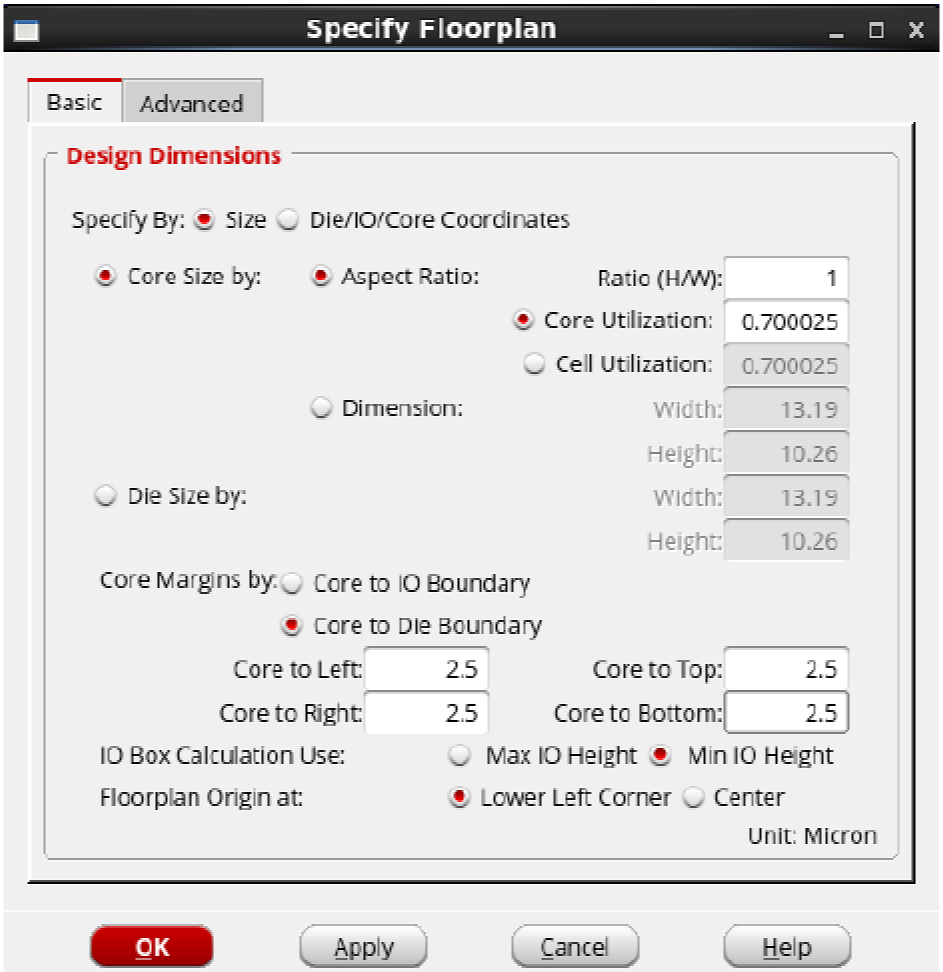


- Netlist: Synthesized Verilog Netlist File[s]
- LEF Files: Library of Components and Physical Data
- Power: Associate VDD to Power Nets and VSS to Ground Nets
- Analysis Configuration: Insert MMMC View Definition File

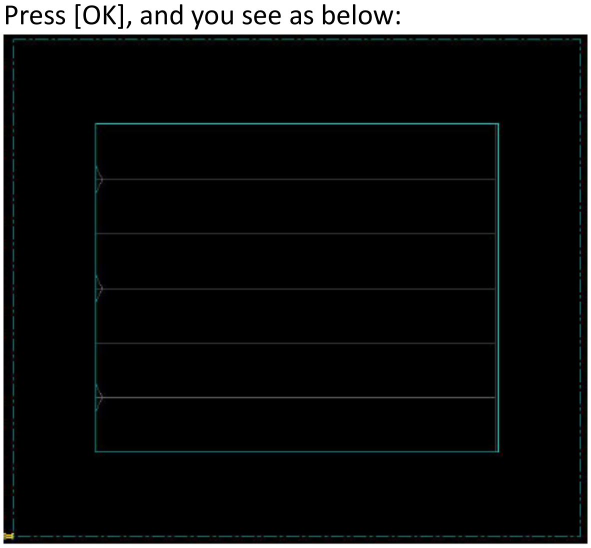
2. Floorplanning the Design

- [Floorplan] – [Specify Floorplan]



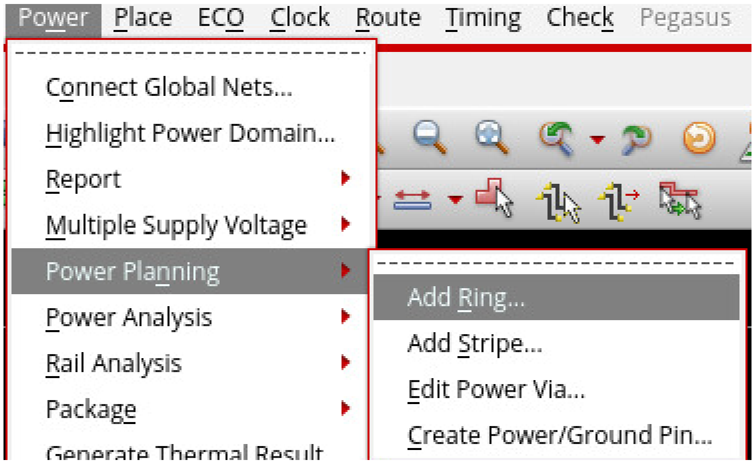


- Aspect Ratio(H/W): 1
- Set Core Margins by "Core to Die Boundary"
- Apply 2.5 micron to each direction as shown
- Press [OK], and you see as below:

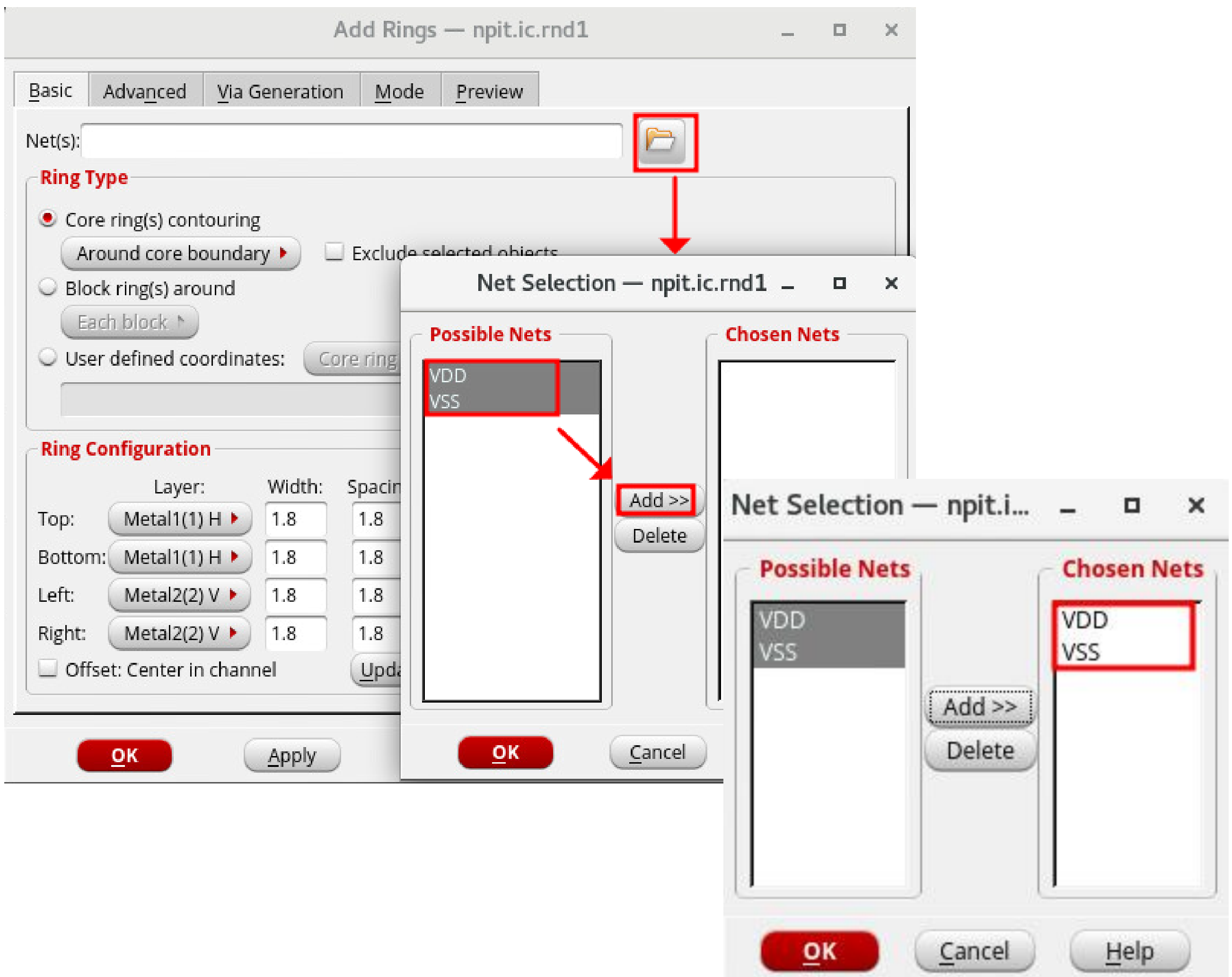


3. Power Planning(1): Add Ring

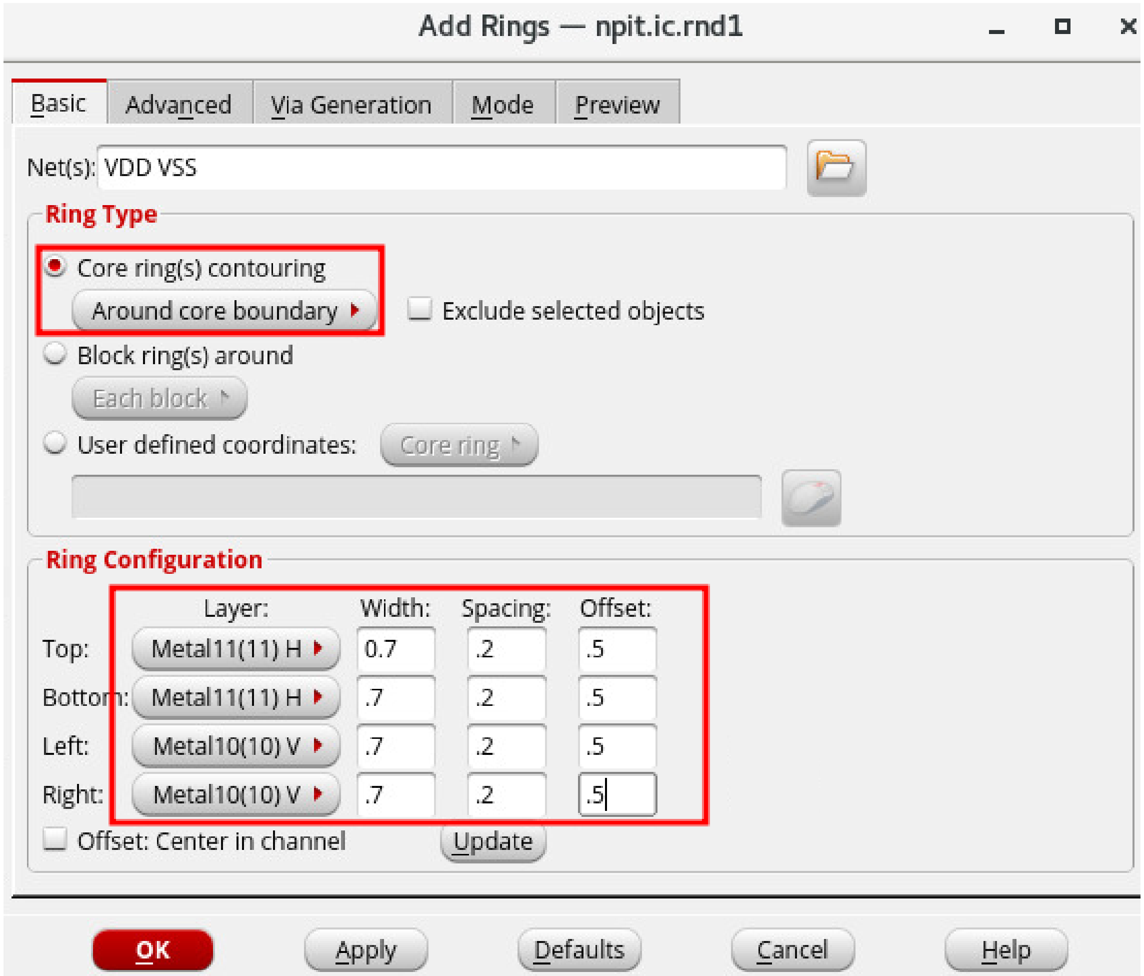
- [Power] - [Power Planning] – [Add Ring]



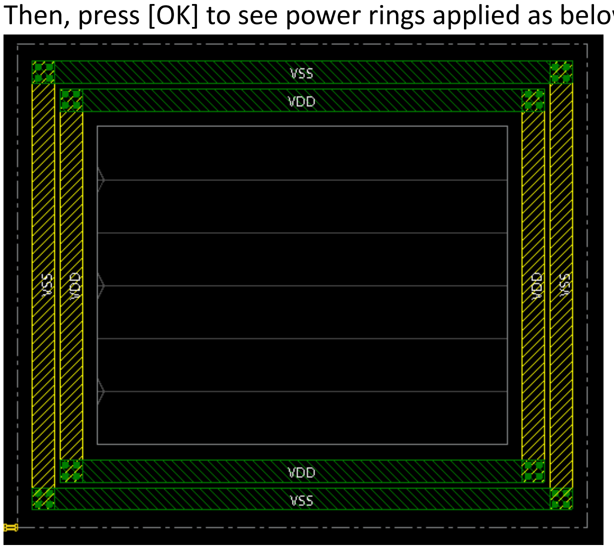
- Select the VDD and VSS nets



- Choose “Around core boundary” to Core ring(s) contouring in Ring Type Filed
- In the Ring Configuration filed, choose Metal11(H) as Top and Bottom Layer and Metal10(V) as Left and Right.
- Set Width 0.7 micron for Width, 0.2 for Spacing and 0.5 for Offset.



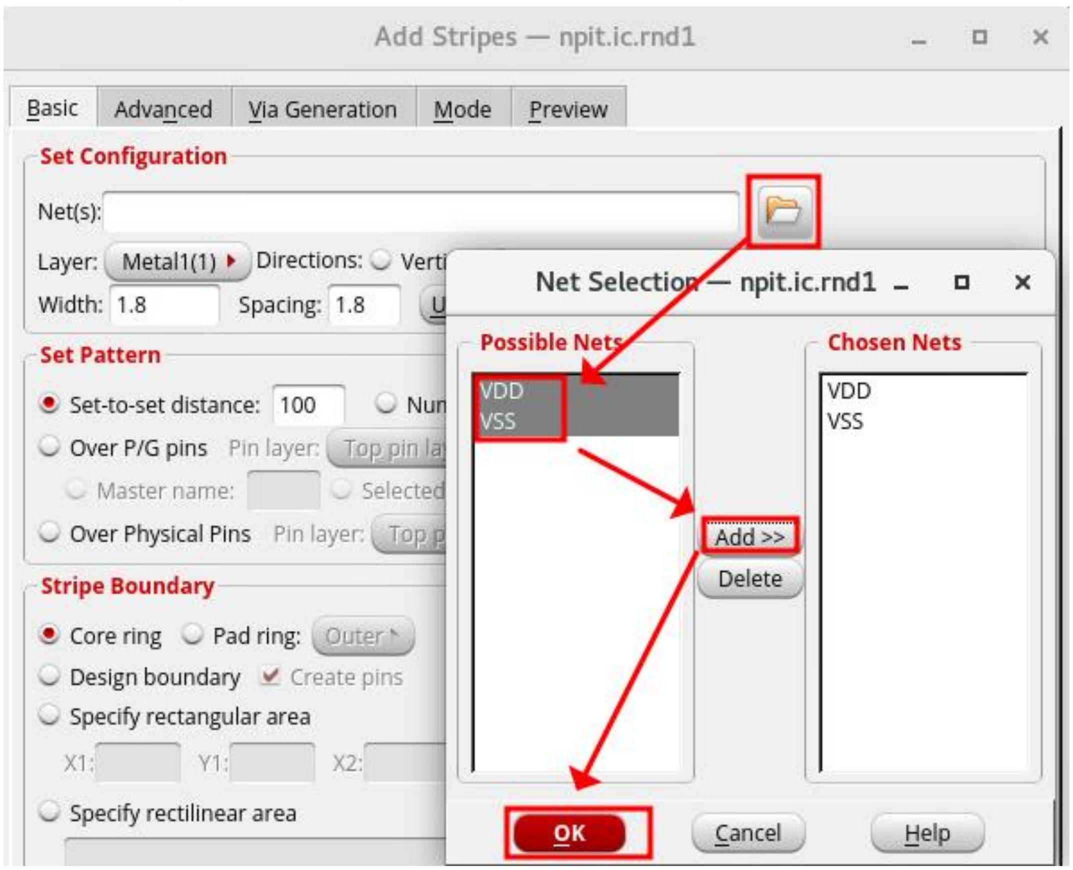
- Then, press [OK] to see power rings applied as below:



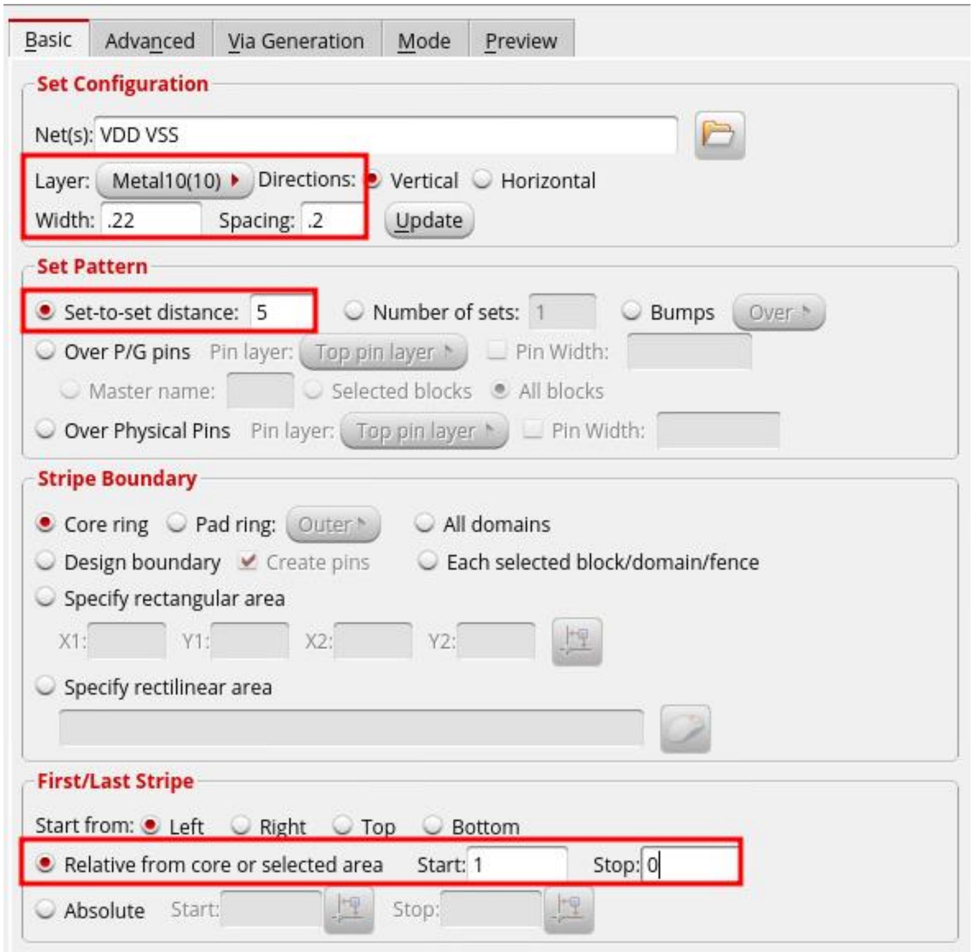
4. Power Planning(2): Add Stripe

- [Power] – [Power Planning] – [Add Stripe]

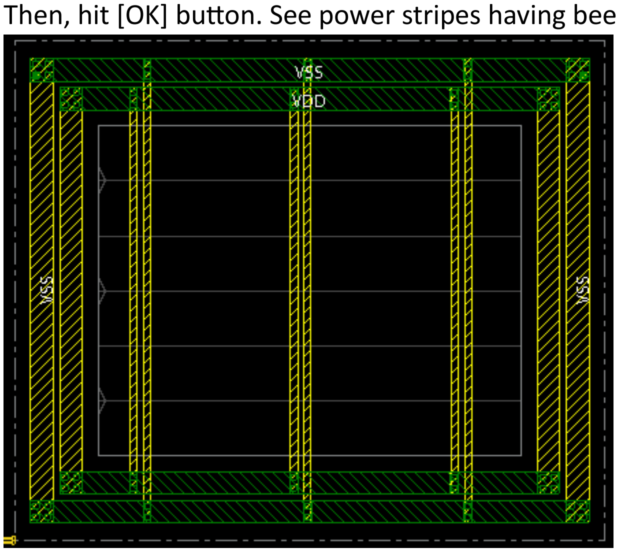
- Add Nets: VDD, VSS



- Set Remaining Fields as below:



- Choose Metal10(10) as vertical layer stripe
- Set 0.22 micron as Width, and 0.2 as Spacing
- Fill in 5 micron as Set-to-set distance
- Set Start to 1n and Stop to 0 as Relative from core or selected area
- Then, hit [OK] button. See power stripes having been made as below:

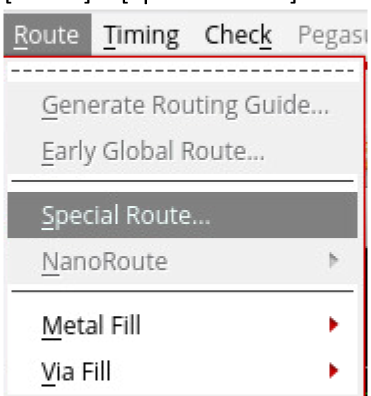


5. Power Planning(3): Power Rail

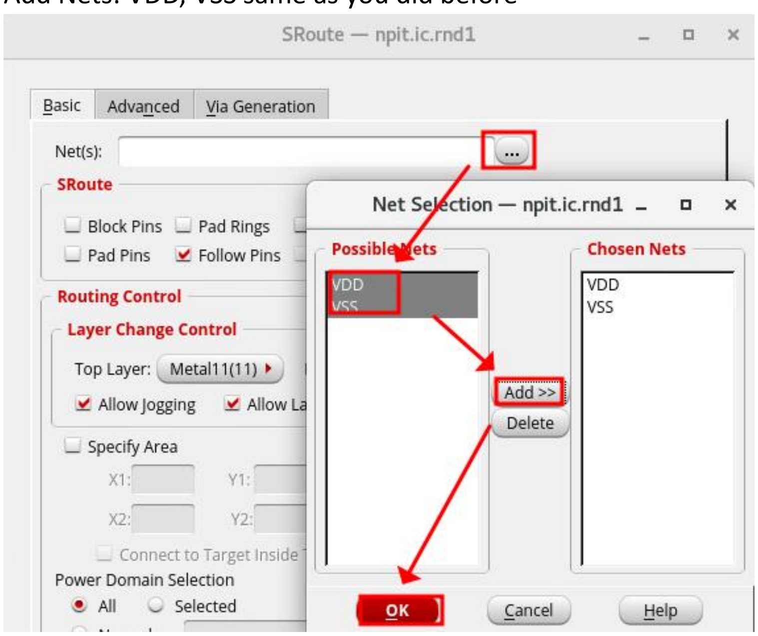
- Associate the global VDD and VDD nets names to the standard cell pin names(a.k.a name mapping) by entering the following commands:
    - innovus @ > connect_global_net VDD -type pg_pin -pin VDD -inst_base_name *
    - innovus @ > connect_global_net VSS -type pg_pin -pin VSS -inst_base_name *

```
@innovus 3> connect_global_net VDD -type pg_pin -pin VDD -inst_base_name *
@innovus 4> connect_global_net VSS -type pg_pin -pin VSS -inst_base_name *
@innovus 5>
```

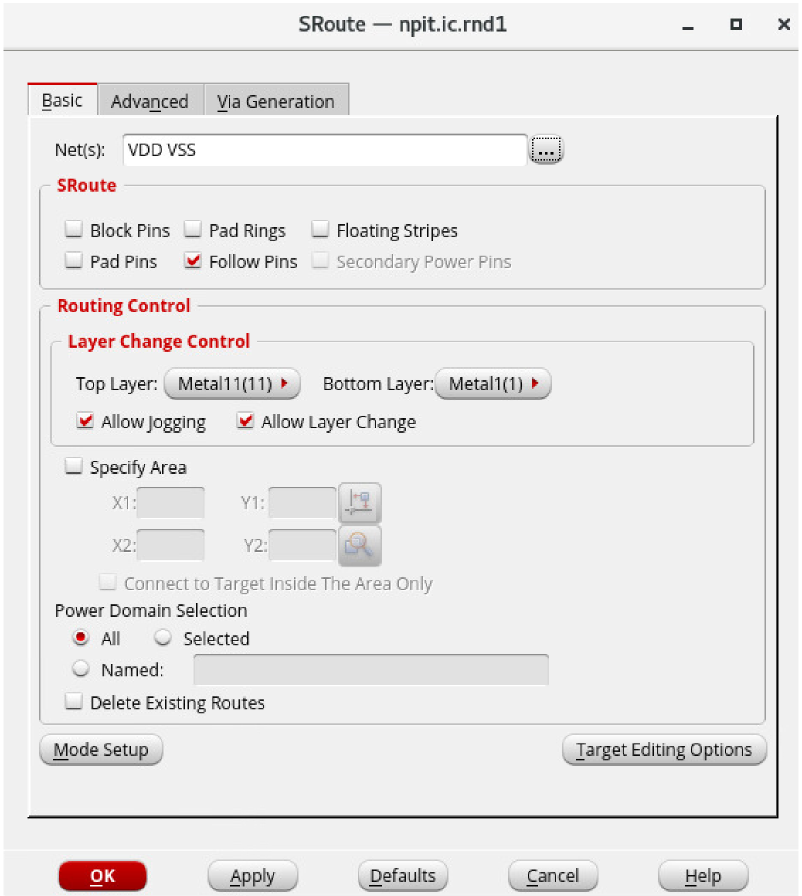
- [Route] – [Special Route]



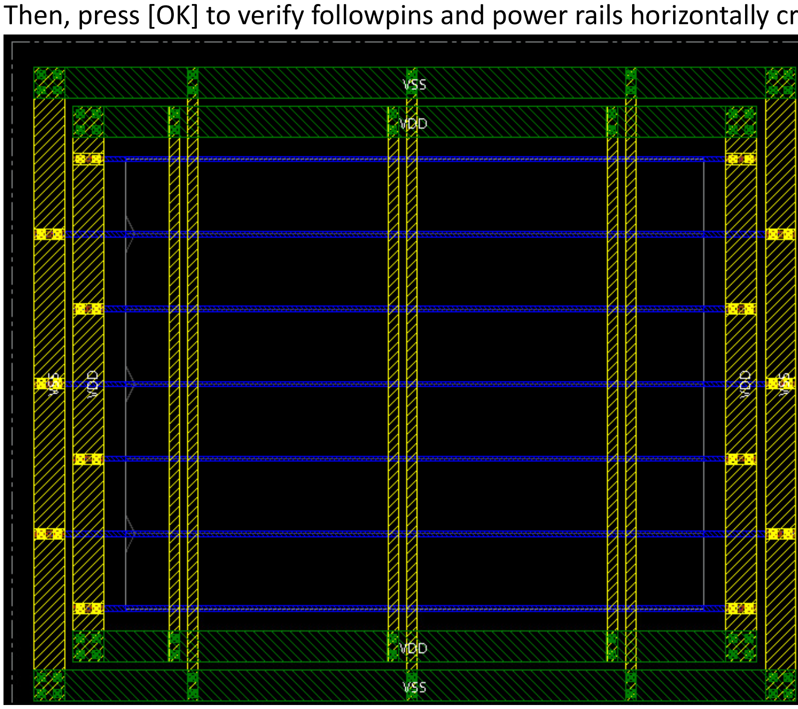
- Add Nets: VDD, VSS same as you did before



- Unselect all pins in the SRoute file EXCEPT “Follow Pins”

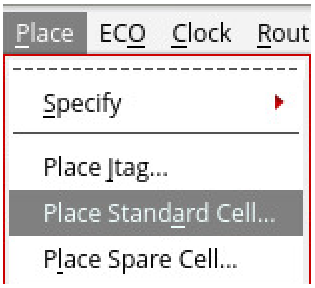


- Then, press [OK] to verify followpins and power rails horizontally created as below:

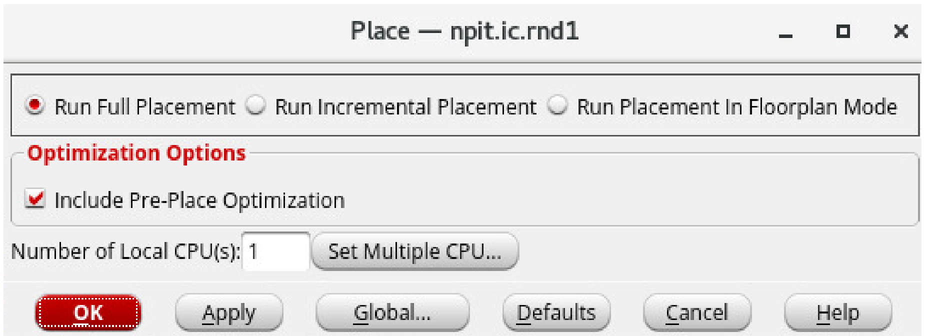


6. Place Standard Cells

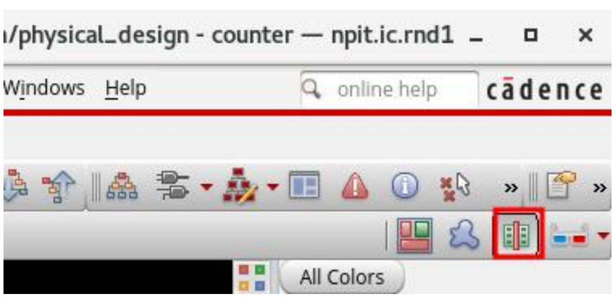
- [Place] – [Place Standard Cells]

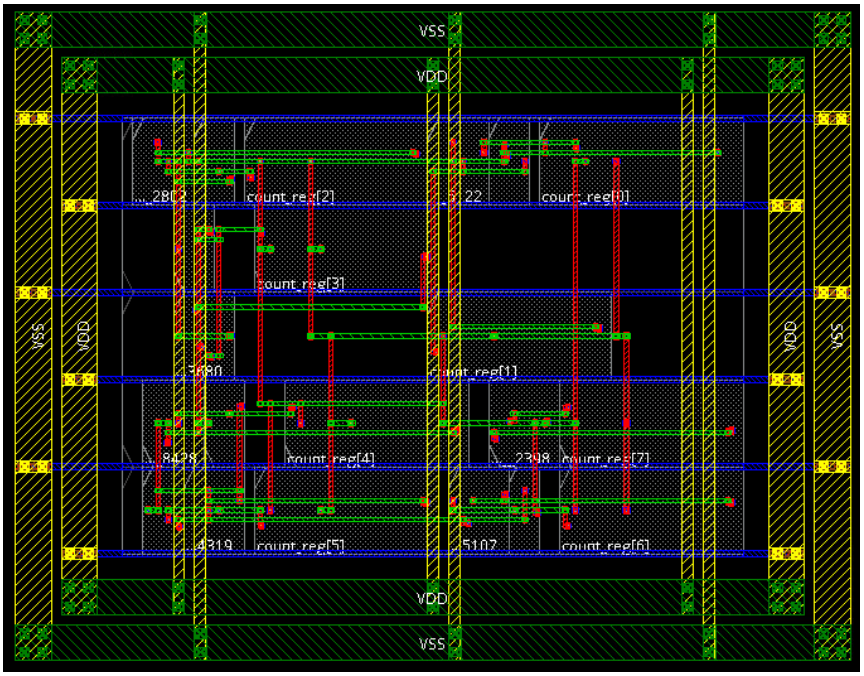


- In the Place field, select “Run Full Placement” and “include Pre-Place Optimization”



- Then, press [OK]
- Press [Physical View Icon] looking like below in the upper right corner to see placed standard cells:





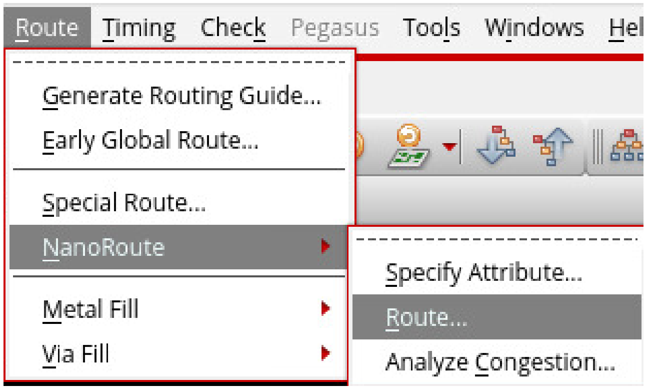
7. Clock Tree Synthesis

- Generate the clock tree spec file constraints from .sdc file
  innovus @ > create_clock_tree_spec

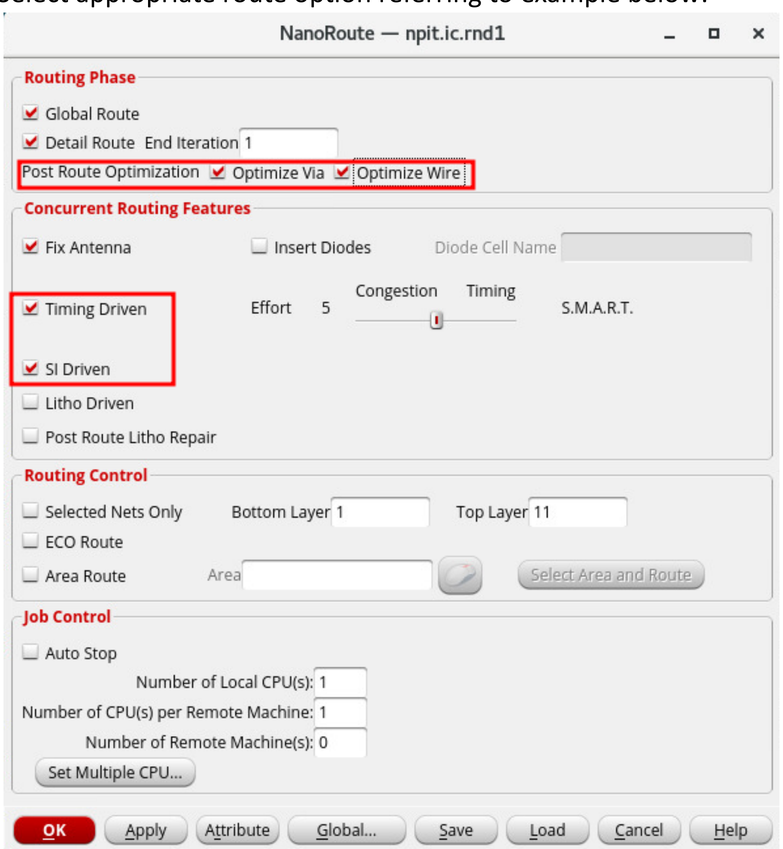
- Create a clock tree
  innovus @ > ccopt_design

8. Route Design

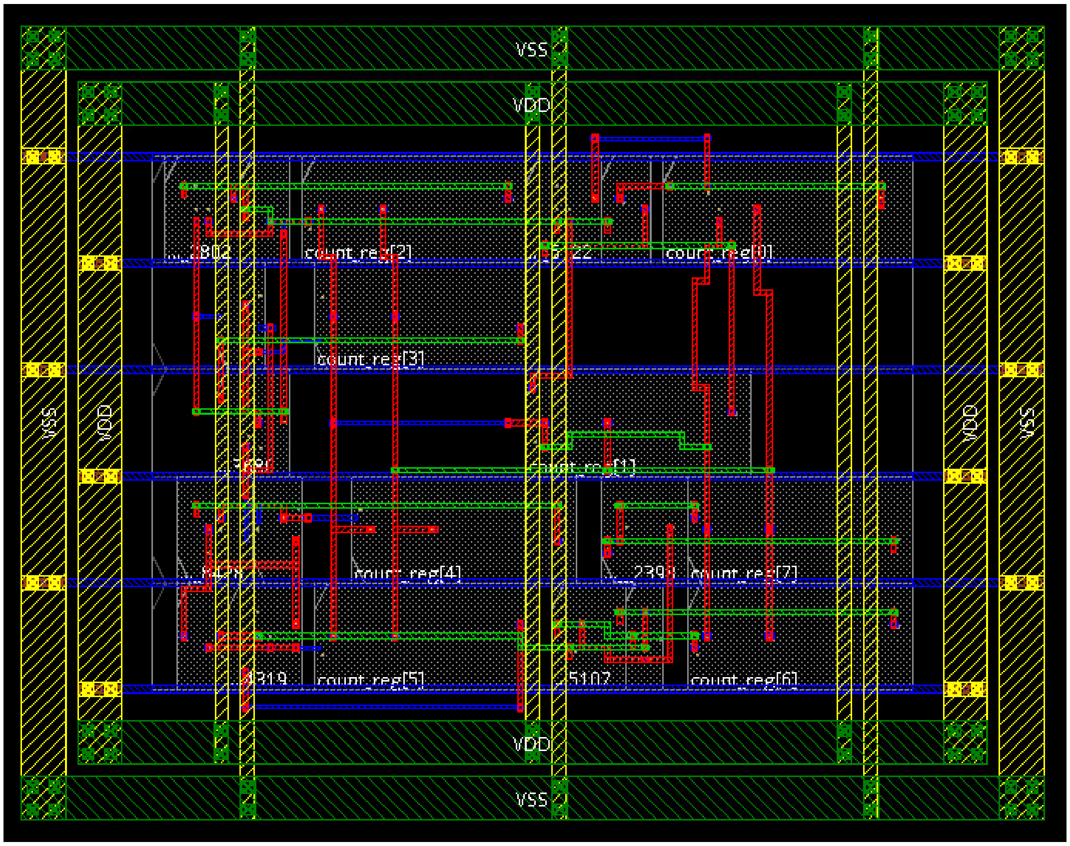
- [Route] – [Nano Route]



- Select appropriate route option referring to example below:

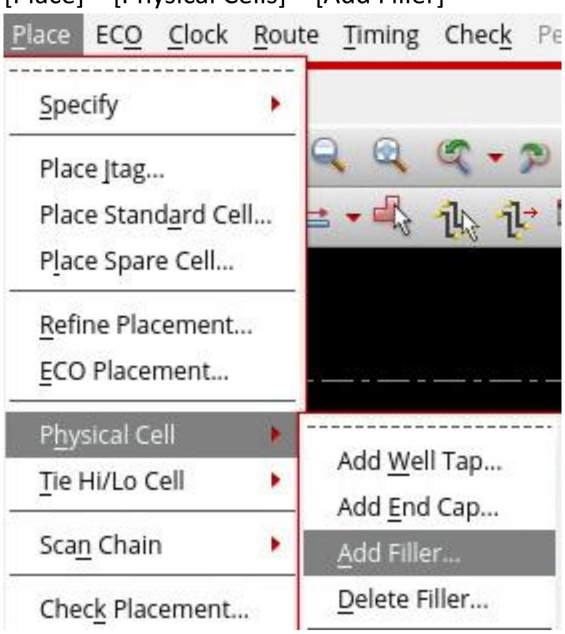


  - Select “Optimize Via” and “Optimize Wire” to reduce DRC error in the Routing Phase
  - Check “Timing Driven” and “SI Driven” for Concurrent Routing Features
  - Then, click [OK] to see the routing result

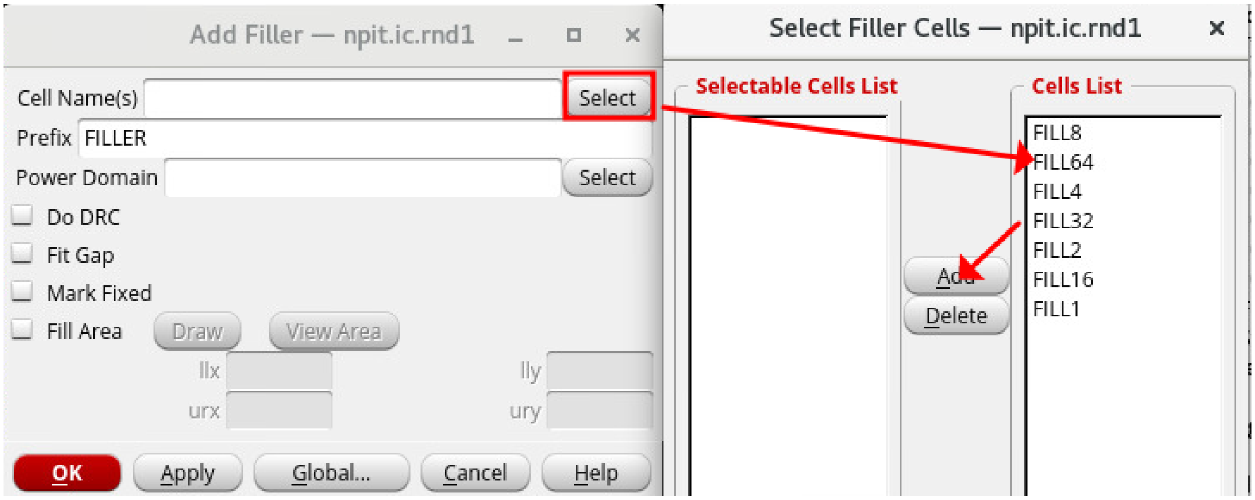


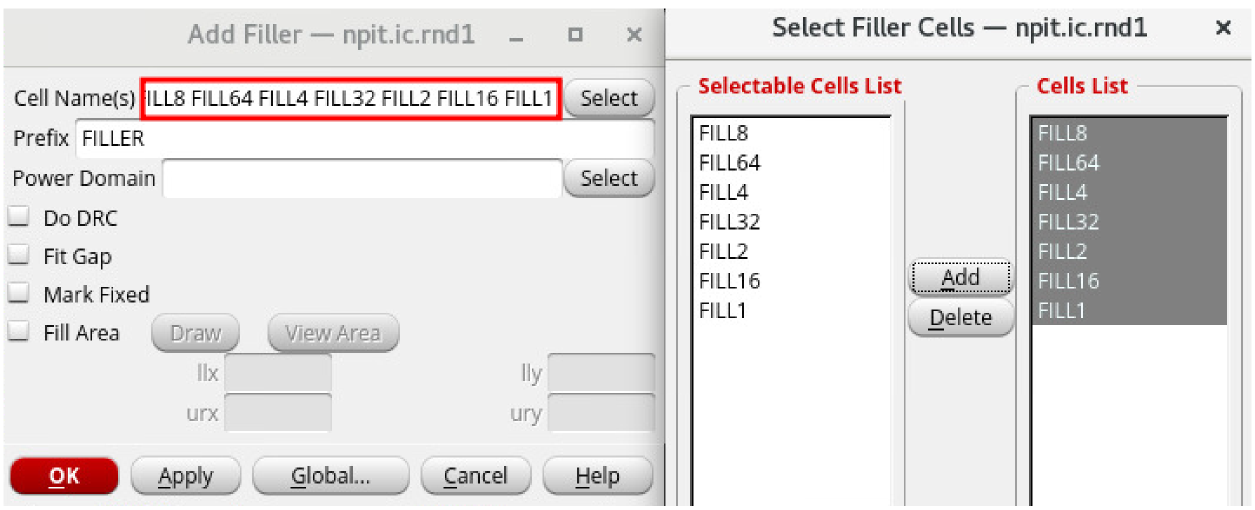
9. Add Filler Cells

- [Place] – [Physical Cells] – [Add Filler]

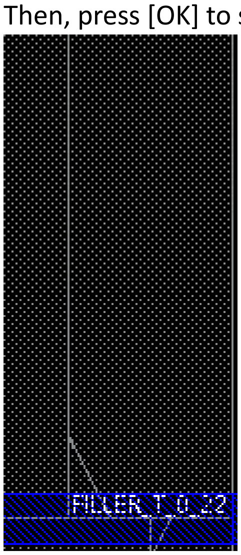


- Fill in Metal Fill Names by hitting [Select] button



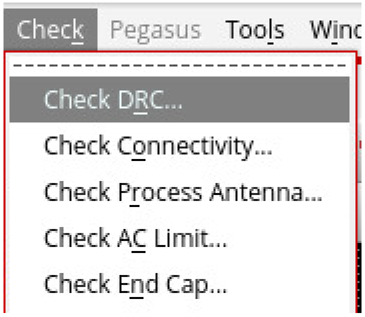


- Then, press [OK] to see the result that the filler cells have been placed

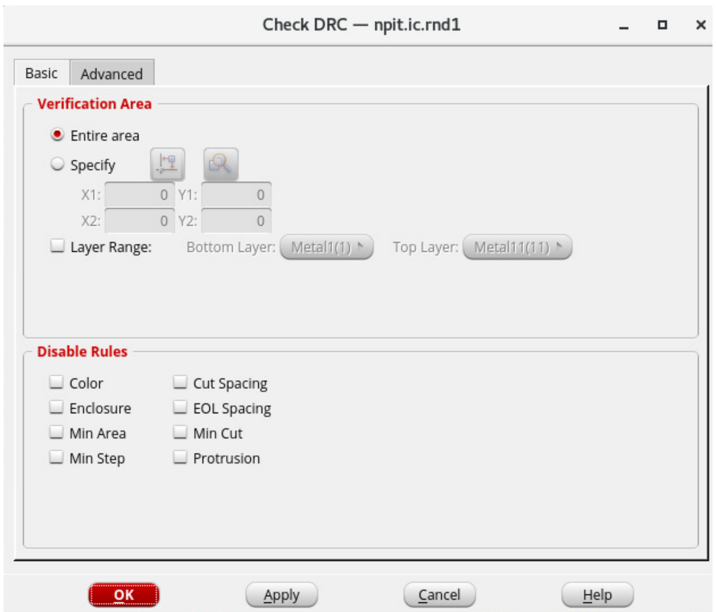


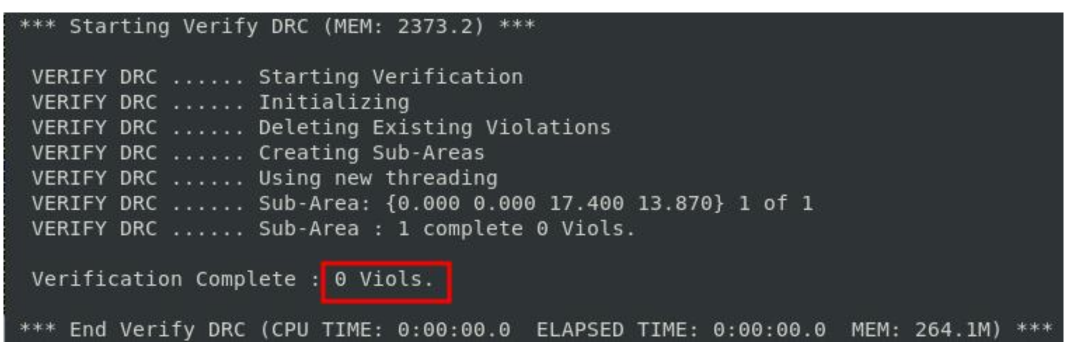
10. DRC

- [Check] – [Check DRC]



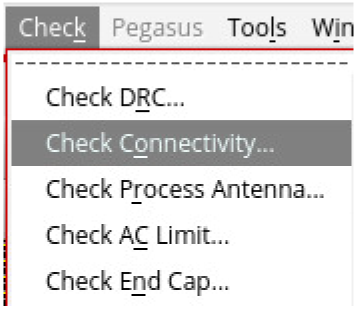
- Press [OK] to see the DRC result in the console terminal



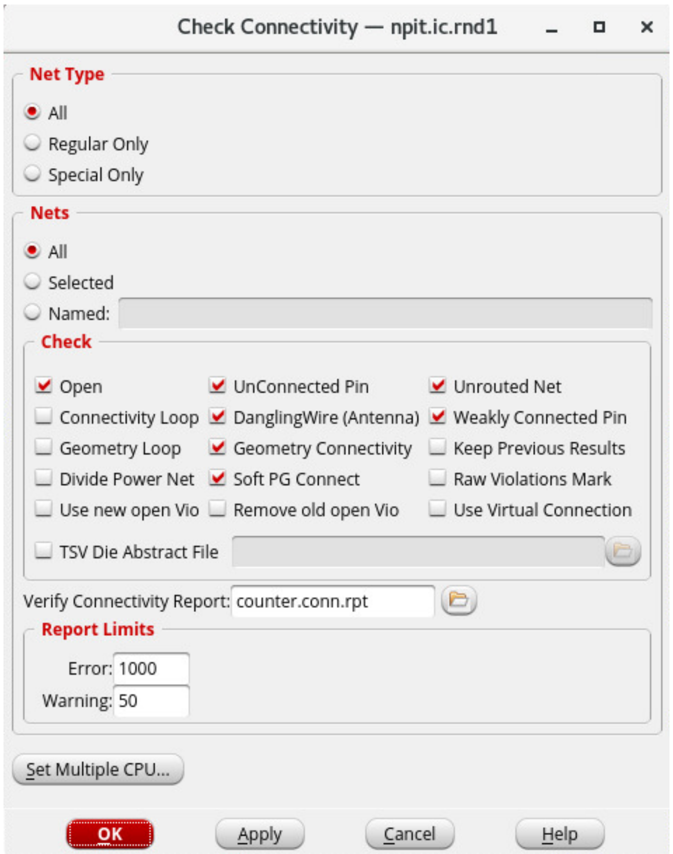


11. Check Connectivity

- [Check] – [Check Connectivity]



- Press [OK] to see the Connectivity result in the console terminal




```
Begin Summary
  Found no problems or warnings.
End Summary

End Time: Sun Aug 18 14:26:15 2024
Time Elapsed: 0:00:00.0

******** End: VERIFY CONNECTIVITY ********
  Verification Complete : 0 Viols.  0 Wrngs.
  (CPU Time: 0:00:00.0  MEM: 0.000M)
```